

# 1. The problem in plain English

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Audience: Charles, re-entering after time away

You do not need any other document open while reading this.

**Binding insight (read early):** The hero shows **outcomes** (draft rate vs pool quality). **Environment** =  $L_{net} = B - D$ . **Advancement** = score ( $S_i, \lambda$ ) then **select** (top K now) — separate from describing the peer environment. See doc 02 and `../BINDING_Selection_is_its_own_step.md`.

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## The one-sentence dissertation idea

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Organizations advance some people and not others. Being surrounded by **strong peers** can **help** you (you learn, you look good by association) and **hurt** you (you are harder to notice, you compete for scarce slots). The dissertation asks whether that **tension** shows up in data — and whether a **simple mechanism** (crowding in the score that ranks who advances) can produce curves that look like the data.

Basketball is one **test bed**: college players who get drafted to the NBA.

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## The basketball “hero” — what we already see in the data

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Think of one row per **player-season**: a college player in a given year, on a given team.

**Question:** Does the chance of eventually being **NBA drafted** relate to the **quality of his teammates**?

We measure teammate quality using box-score performance (points per minute, adjusted within season). Crucially, we use **leave-one-out** quality: average teammate performance **excluding the player himself**, so he is not counted in his own pool average.

Call that number **pool quality (leave-one-out)**. Higher means better teammates around him.

**What we plot:** Split all player-seasons into 16 equal-sized bins from lowest to highest pool quality. For each bin, compute the **fraction who were ever drafted**. Connect the dots.

**What we see (qualitative shape):**

- Draft rate **rises** as pool quality increases through the middle bins.
- Rate **levels off** in high-but-not-top bins.
- In the **very top bin only**, rate **drops** — an inverted-U **tail**, not necessarily a symmetric hill.

That plot is the **hero**. It is a **stylized fact** — something real in the NCAA panel under fixed rules (which seasons, which minutes filter, which binning). It is **not** yet a story about why NBA teams behave that way.

**First step — two axes, two curves (real data):** Before the hero is “surprising,” check the obvious axis. Bin the **same** player-seasons by **own college performance** (same ppm stat, z-scored within season) instead of teammate pool quality. Draft rate then rises **monotonically** from the bottom talent bin to the top (~0% → ~7% on the locked July 2026 sample). That is the **talent baseline** — talent matters, as it should. The **hero** bins on **pool quality** instead and gets a **different** shape: rise, plateau, then a **dip in the top pool-quality bin only**. The phenomenon is the **second** curve, not draft itself. We are **not** claiming the NBA ignores talent; we are showing that **peer environment** has its own non-monotone pattern once you look at it on its own axis.

**Important:** The hero shows **who got drafted** vs pool quality. It does **not** tell you whether the bend comes from peer effects on **development** or from **congestion in the ranking/score** that feeds who gets selected. That split is Layer B (doc 02): environment (B – D) vs **score then select**.

**Sample size (locked July 2026):** about 62,000 player-seasons; about 1,100 drafted.

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## Why this matters for the “simplified model” push

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Alex’s scientific arc (paraphrased): show the **phenomenon**, then show a **minimal generative rule** where **talent alone is not enough** — you need something like **congestion in the score** — then later add richer measurements and predictions.

You are **not** being asked, in the next few days, to:

- perfectly reproduce every bin of the hero from simulation;
- estimate separate “benefit” and “congestion” functions from one chart;
- unify Army, basketball, and tenure in one model.

You **are** being asked to hold a clear line:

1. **Here are the two empirical curves** — talent baseline (monotone) and hero (pool quality with tail dip); both on real NCAA data under the locked spec.
2. **Here is a disciplined artificial league** where **scoring** purely on talent gives a different curve than scoring on talent **minus** a congestion penalty — same winner rule (top K), side by side, honest axes, explicit limits on what we claim.

That is the simplified model **deliverable** for basketball v1. Slide summary: `Model.pdf`.

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Words you will see again (defined once here)

| Term                   | Plain meaning                                                                                                                                                                                                                |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Panel                  | The main analysis table of <b>real</b> college player-seasons — one row per athlete × season × team, with own stats, leave-one-out teammate quality, and draft outcome; we <b>filter</b> it, then <b>bin</b> it for the hero |
| Hero                   | The binned draft-rate plot vs leave-one-out pool quality                                                                                                                                                                     |
| Talent baseline        | Same panel binned by <b>own</b> performance instead of pool quality; draft rate rises monotonically (sanity check before the hero)                                                                                           |
| Pool quality (LOO)     | How good your teammates are on paper, with you removed from the average                                                                                                                                                      |
| Draft rate             | Share of players in a bin who were ever NBA drafted                                                                                                                                                                          |
| Stylized fact          | A robust pattern in data under stated rules — not yet causal proof                                                                                                                                                           |
| Generative model / sim | A computer-made fake league with explicit draft rules                                                                                                                                                                        |
| Congestion             | “Too many good peers” — harder to stand out or win a scarce slot                                                                                                                                                             |
| BINDING                | Charles-locked rule: keep <b>L_net</b> (environment) separate from advancement; keep <b>score</b> ( <b>S_i</b> ) separate from <b>select</b> (top K / later noise)                                                           |

Emotional checkpoint

If the only thing you take from this page is:

*Good players get drafted more than weak ones (talent baseline). But among teammate environments, the very best pools do not have the highest draft odds — and we have a fixed hero plot that shows that.*

—you have enough to read document **02** (three kinds of model) without guilt about everything else in the repo.

Next: [02\\_Three\\_Kinds\\_of\\_Model.md](#)